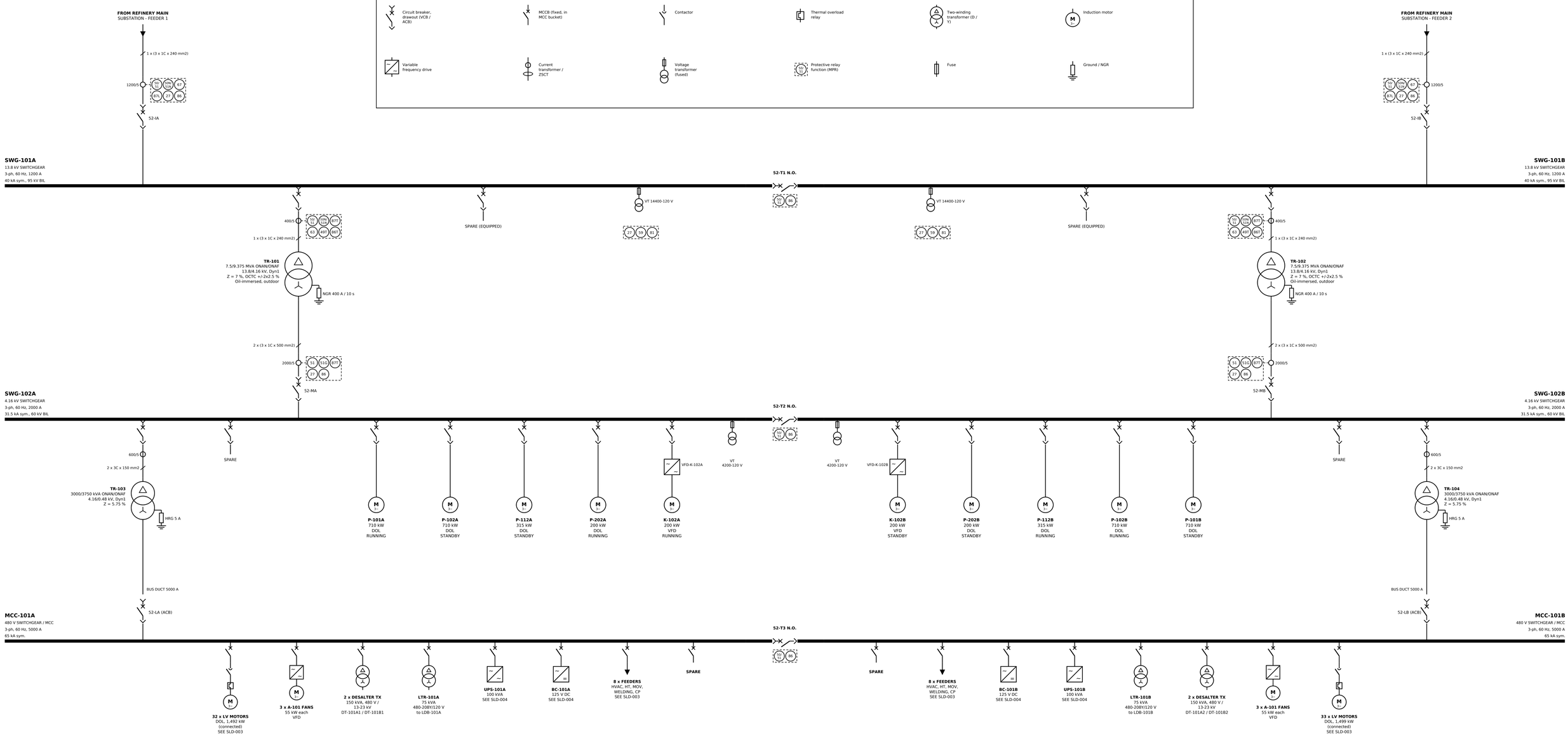


MAXIMUM DEMAND SUMMARY (MD = C + 0.3 I + 0.1 S)

BUS	C kW	I kW	S kW	MD kW	MD kvar	MD kVA	PF
MCC-101A	1,084	127	559	1,178	643	1,342	0.88
MCC-101B	1,080	124	575	1,175	634	1,335	0.88
SWG-102A (MV motors)	985	0	918	1,077	528	1,199	0.90
SWG-102A (incl. LV)				2,266	1,197	2,563	0.88
SWG-102B (MV motors)	918	0	985	1,017	534	1,148	0.89
SWG-102B (incl. LV)				2,203	1,193	2,505	0.88
UNIT TOTAL at 13.8 kV				**4,505**	**2,533**	**5,168**	**0.87**

MAIN EQUIPMENT RATINGS

TAG	DESCRIPTION	RATING	SC / LOAD
SWG-101A/B	13.8 kV metal-clad, VCB	1200 A	40 kA
TR-101/102	7.5/9.375 MVA 13.8/4.16 kV	Z 7 %	68 % MD
SWG-102A/B	4.16 kV metal-clad, VCB, AR	2000 A	31.5 kA
TR-103/104	3000/3750 kVA 4.16/0.48 kV	Z 5.75 %	89 % MD
MCC-101A/B	480 V LV swgr + MCC	5000 A	65 kA
UPS-101A/B	2 x 100 kVA, 30 min	VRLA	350 Ah

FAULT LEVELS (max. case)

BUS	Ik" kA	ip kA	RATED kA
13.8 kV	32.2	83.0	40
4.16 kV	15.9	40.5	31.5
0.48 kV	52.4	123.3	65

MOTOR STARTING (largest DOL motor):
P-101A 710 kW DOL at 4.16 kV:
bus dip 5.3 % (limit 10 %),
terminal dip 5.8 % (limit 15 %)

NOTES

1. Normal operation: both incomers closed, bus ties NORMALLY OPEN (N.O.) at 13.8 kV, 4.16 kV and 480 V.
2. Each transformer sized for the total unit maximum demand (one out of service, tie closed): ONAN >= MD, ONAF >= 1.25 x MD (25 % future). See CFU-000-EL-CAL-001.
3. Tie breakers: 2-out-of-3 electrical/key interlock per board (no paralleling of transformers); auto-transfer (open transition) on loss of incomer, with residual-voltage supervision (27R).
4. 4.16 kV system low-resistance grounded (400 A); 480 V high-resistance grounded (5 A, alarm).
5. Pump A/B (2 x 100 %) are always fed from opposite bus sections; the normally running unit of each pair is selected to balance the buses (load list CFU-000-EL-LDL-001).
6. No emergency diesel generator: essential loads (DCS/SIS/F&G/telecom) on 2 x 100 % UPS, 30 min autonomy; unit fails safe on total power loss. See SLD-004 note 1.
7. Detailed MV / LV / UPS SLDs: CFU-000-EL-SLD-002 / -003 (2 sheets) / -004.
8. Short-circuit ratings: symmetrical interrupting kA; utility 31.5 kA at 13.8 kV (basis).

REV	DATE	DESCRIPTION
A	2026-10-02	Issued for review (FEED)

GENERIC REFINERY - GULF COAST
100 KBPSD CRUDE & VACUUM DISTILLATION UNIT

OVERALL KEY SINGLE LINE DIAGRAM

13.8 / 4.16 / 0.48 KV DISTRIBUTION - SUBSTATION SS-100

DISCIPLINE	STAGE	SCALE	SHEET
ELECTRICAL	FEED	NTS	1 OF 1

DRAWN	CHECKED	APPROVED	DATE
CLAUDE CODE	-	-	2026-10-02

DRAWING NUMBER: **CFU-000-EL-SLD-001** REV: **A**